

Tarea 1

SP-1637 Análisis de Supervivencia — IIC-2026

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Carga de datos

```
pacman::p_load(haven, dplyr, survival, survminer, patchwork)

datos <- read_dta("encfec2010.dta") |>
  filter(region == 1 | region == 8)

summary(datos)
```

region	zona	r201a	r201b
Min. :1.000	Min. :1.000	Min. : 1.00	Min. : 1.000
1st Qu.:1.000	1st Qu.:1.000	1st Qu.: 7.00	1st Qu.: 7.000
Median :1.000	Median :1.000	Median :14.00	Median : 9.000
Mean :1.709	Mean :1.158	Mean :14.94	Mean : 7.927
3rd Qu.:1.000	3rd Qu.:1.000	3rd Qu.:22.00	3rd Qu.:10.000
Max. :8.000	Max. :2.000	Max. :31.00	Max. :99.000

r201c	r202	r204	ednivel	r209
Min. :1930	Min. :15.00	Min. :1.000	Min. : 0.00	Min. :1.000
1st Qu.:1960	1st Qu.:21.00	1st Qu.:1.000	1st Qu.:10.00	1st Qu.:1.000

Median :1977	Median :32.00	Median :1.000	Median :20.00	Median :1.000
Mean :1973	Mean :36.52	Mean :1.498	Mean :24.55	Mean :1.961
3rd Qu.:1988	3rd Qu.:50.00	3rd Qu.:2.000	3rd Qu.:40.00	3rd Qu.:2.000
Max. :1995	Max. :80.00	Max. :2.000	Max. :99.00	Max. :9.000

r215	r216m	r216a	id
Min. :1.000	Min. : 1.00	Min. :1949	Min. : 1
1st Qu.:1.000	1st Qu.: 4.00	1st Qu.:1980	1st Qu.: 562
Median :2.000	Median : 8.00	Median :1995	Median :1100
Mean :1.754	Mean :21.93	Mean :2868	Mean :1188
3rd Qu.:2.000	3rd Qu.:12.00	3rd Qu.:2005	3rd Qu.:1675
Max. :9.000	Max. :99.00	Max. :9999	Max. :3195
	NAs :635	NAs :635	

```
table(datos$region)
```

```
1      8
1554 175
```

Ejercicio 1

```
datos <- datos |>
mutate(
  r201b = as.numeric(na_if(r201b, 99)),
  r201c = as.numeric(na_if(r201c, 9999)),
  r216m = if_else(r216m %in% c(98, 99), NA, as.numeric(r216m)),
  r216a = if_else(r216a %in% c(9998, 9999), NA, as.numeric(r216a)),
  evento = case_when(
    r215 == 1 ~ 0,
    r215 %in% c(2, 3) ~ 1
  ),
  tiempo_union = case_when(
    evento == 0 ~ as.numeric(r202),
    evento == 1 ~ ((r216a - r201c) * 12 + (r216m - r201b)) / 12
  )
)

summary(datos$r201b)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NAs
1.000	7.000	9.000	7.874	10.000	12.000	1

```
summary(datos$r216m)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NAs
1.000	4.000	7.000	6.814	10.000	12.000	815

```
table(datos$r215, datos$evento, useNA = "ifany")
```

	0	1	<NA>
1	634	0	0
2	0	892	0
3	0	202	0

9 0 0 1

```
summary(datos$tiempo_union)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NAs
9.00	18.00	21.33	23.61	25.94	79.00	201

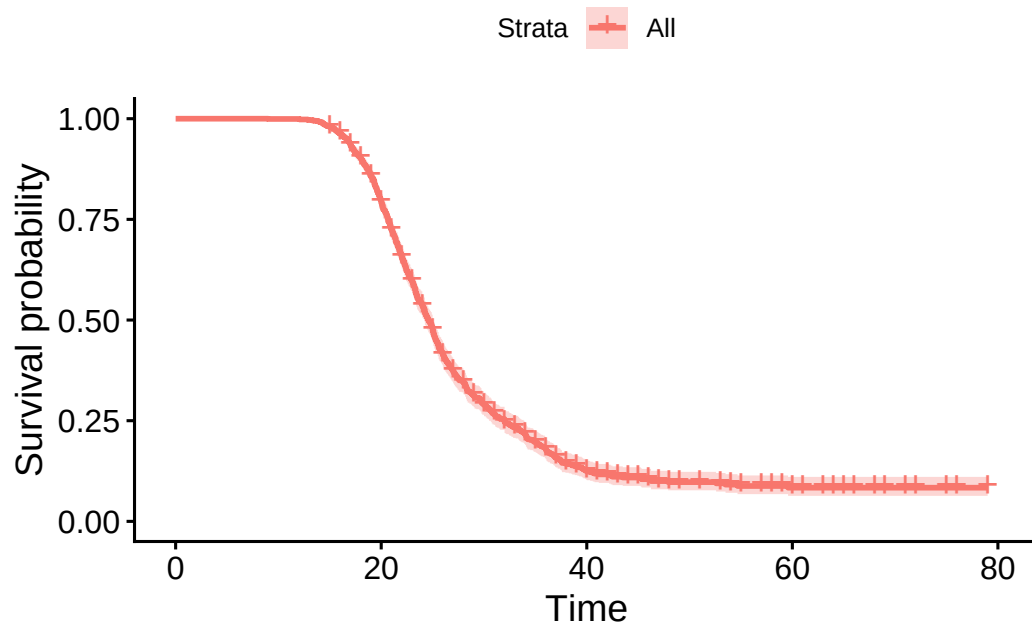
Ejercicio 2

Metropolitana y Pacífico Central

```
so.1 <- Surv(time = datos$tiempo_union, event = datos$evento)
fit1 <- survfit(so.1 ~ 1)
summary(fit1)$table
```

records	n.max	n.start	events	rmean	se(rmean)
1528.0000000	1528.0000000	1528.0000000	894.0000000	30.0594254	0.5913513
median	0.95LCL	0.95UCL			
24.5833333	24.1666667	25.0833333			

```
ggsurvplot(fit1, data = datos)
```



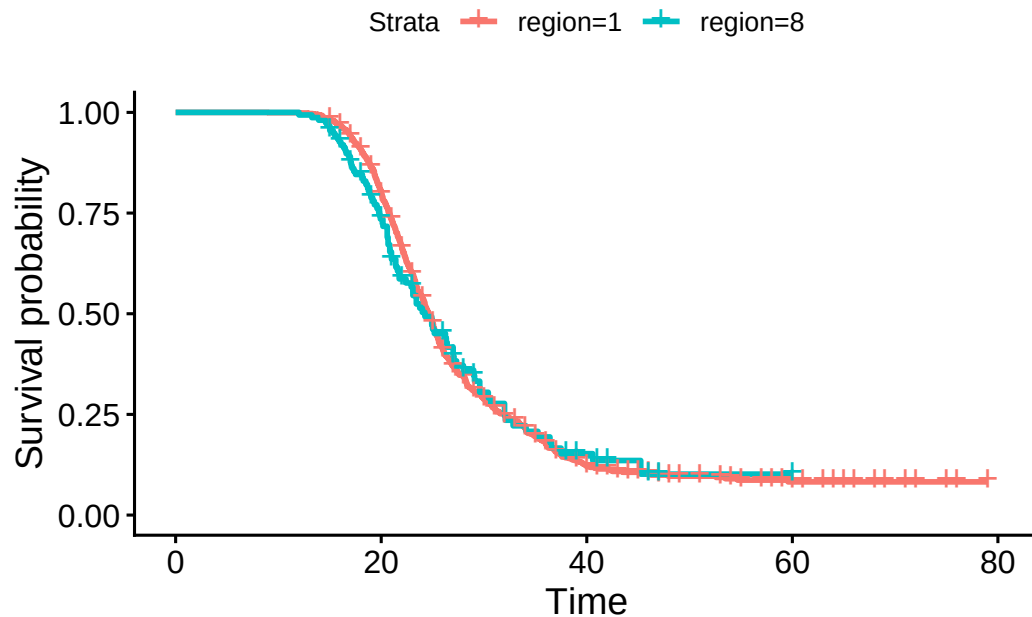
La edad mediana a la primera unión es de aproximadamente 25 años para las personas que viven en la región Pacífico Central y Metropolitana

La proporción de personas que nunca se unen es de aproximadamente 0,1 o un 10 % de las personas de las regiones Pacífico Central y Metropolitana que nunca se unen

Estratificando por región

```
so.2 <- Surv(time = datos$tiempo_union, event = datos$evento)
fit2 <- survfit(so.2 ~ region, data = datos)

ggsurvplot(fit2, data = datos)
```



La edad mediana a la primera unión es de aproximadamente 25 años para las personas que viven en la región Pacífico Central como para las que habitan en la región Metropolitana

La proporción de personas que nunca se unen es de aproximadamente 0,1 o un 10 % en la región Pacífico Central así como en la región Metropolitana

Ejercicio 3

```
summary(fit1)
```

Call: survfit(formula = so.1 ~ 1)

201 observations deleted due to missingness

time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI
9.0	1528	1	0.9993	0.000654	0.9981	1.000
12.0	1527	1	0.9987	0.000925	0.9969	1.000
12.1	1526	1	0.9980	0.001132	0.9958	1.000
12.8	1525	1	0.9974	0.001307	0.9948	1.000
12.9	1524	1	0.9967	0.001461	0.9939	1.000
13.2	1523	1	0.9961	0.001600	0.9929	0.999
13.3	1522	1	0.9954	0.001728	0.9920	0.999
13.5	1521	1	0.9948	0.001846	0.9912	0.998
13.8	1520	1	0.9941	0.001958	0.9903	0.998
13.8	1519	1	0.9935	0.002063	0.9894	0.998
13.9	1518	1	0.9928	0.002163	0.9886	0.997

14.0	1517	1	0.9921	0.002258	0.9877	0.997
14.1	1516	1	0.9915	0.002350	0.9869	0.996
14.2	1515	3	0.9895	0.002604	0.9844	0.995
14.3	1512	3	0.9876	0.002835	0.9820	0.993
14.4	1509	3	0.9856	0.003047	0.9796	0.992
14.6	1506	3	0.9836	0.003245	0.9773	0.990
14.7	1503	2	0.9823	0.003370	0.9757	0.989
14.8	1501	1	0.9817	0.003431	0.9750	0.988
14.8	1500	1	0.9810	0.003491	0.9742	0.988
15.0	1499	3	0.9791	0.003663	0.9719	0.986
15.2	1410	2	0.9777	0.003787	0.9703	0.985
15.2	1408	1	0.9770	0.003848	0.9695	0.985
15.3	1407	1	0.9763	0.003907	0.9687	0.984
15.4	1406	2	0.9749	0.004023	0.9670	0.983
15.5	1404	4	0.9721	0.004245	0.9638	0.980
15.6	1400	3	0.9700	0.004403	0.9614	0.979
15.7	1397	2	0.9686	0.004504	0.9599	0.978
15.8	1395	3	0.9666	0.004653	0.9575	0.976
15.8	1392	2	0.9652	0.004748	0.9559	0.975
15.9	1390	2	0.9638	0.004842	0.9543	0.973
16.0	1388	2	0.9624	0.004934	0.9528	0.972
16.1	1335	2	0.9610	0.005030	0.9511	0.971
16.2	1333	2	0.9595	0.005125	0.9495	0.970
16.2	1331	3	0.9573	0.005264	0.9471	0.968
16.3	1328	4	0.9545	0.005442	0.9439	0.965
16.4	1324	1	0.9537	0.005485	0.9431	0.965
16.5	1323	3	0.9516	0.005613	0.9406	0.963
16.6	1320	2	0.9501	0.005696	0.9390	0.961
16.7	1318	2	0.9487	0.005778	0.9374	0.960
16.8	1316	5	0.9451	0.005977	0.9334	0.957
16.8	1311	5	0.9415	0.006167	0.9295	0.954
16.9	1306	3	0.9393	0.006278	0.9271	0.952
17.0	1303	8	0.9336	0.006563	0.9208	0.947
17.1	1229	5	0.9298	0.006752	0.9166	0.943
17.2	1224	3	0.9275	0.006863	0.9141	0.941
17.2	1221	4	0.9244	0.007006	0.9108	0.938
17.3	1217	6	0.9199	0.007215	0.9059	0.934
17.4	1211	3	0.9176	0.007316	0.9034	0.932
17.5	1208	3	0.9153	0.007415	0.9009	0.930
17.6	1205	2	0.9138	0.007480	0.8993	0.929
17.7	1203	4	0.9108	0.007608	0.8960	0.926
17.8	1199	1	0.9100	0.007639	0.8952	0.925
17.8	1198	5	0.9062	0.007794	0.8911	0.922
17.9	1193	3	0.9039	0.007885	0.8886	0.920
18.0	1190	4	0.9009	0.008003	0.8853	0.917
18.1	1125	4	0.8977	0.008133	0.8819	0.914
18.2	1121	5	0.8937	0.008292	0.8776	0.910
18.2	1116	5	0.8897	0.008446	0.8733	0.906
18.3	1111	5	0.8857	0.008596	0.8690	0.903
18.4	1106	2	0.8841	0.008654	0.8673	0.901
18.5	1104	4	0.8809	0.008770	0.8639	0.898
18.6	1100	6	0.8761	0.008939	0.8587	0.894
18.7	1094	2	0.8745	0.008994	0.8570	0.892
18.8	1092	6	0.8697	0.009156	0.8519	0.888

18.8	1086	4	0.8665	0.009261	0.8485	0.885
18.9	1082	9	0.8593	0.009491	0.8409	0.878
19.0	1073	4	0.8560	0.009589	0.8375	0.875
19.1	1024	3	0.8535	0.009670	0.8348	0.873
19.2	1021	3	0.8510	0.009749	0.8321	0.870
19.2	1018	6	0.8460	0.009905	0.8268	0.866
19.3	1012	11	0.8368	0.010178	0.8171	0.857
19.4	1001	7	0.8310	0.010344	0.8109	0.851
19.5	994	5	0.8268	0.010460	0.8065	0.848
19.6	989	8	0.8201	0.010639	0.7995	0.841
19.7	981	6	0.8151	0.010769	0.7943	0.836
19.8	975	5	0.8109	0.010875	0.7899	0.833
19.8	970	9	0.8034	0.011059	0.7820	0.825
19.9	961	5	0.7992	0.011159	0.7776	0.821
20.0	956	10	0.7908	0.011351	0.7689	0.813
20.1	911	7	0.7848	0.011494	0.7626	0.808
20.2	904	11	0.7752	0.011709	0.7526	0.799
20.2	893	2	0.7735	0.011747	0.7508	0.797
20.3	891	4	0.7700	0.011822	0.7472	0.794
20.4	887	5	0.7657	0.011913	0.7427	0.789
20.5	882	6	0.7605	0.012021	0.7373	0.784
20.6	876	7	0.7544	0.012142	0.7310	0.779
20.7	869	11	0.7448	0.012325	0.7211	0.769
20.8	858	3	0.7422	0.012373	0.7184	0.767
20.8	855	9	0.7344	0.012514	0.7103	0.759
20.9	846	7	0.7283	0.012619	0.7040	0.753
21.0	839	6	0.7231	0.012707	0.6986	0.748
21.1	784	6	0.7176	0.012809	0.6929	0.743
21.2	778	6	0.7121	0.012908	0.6872	0.738
21.2	772	4	0.7084	0.012972	0.6834	0.734
21.3	768	6	0.7028	0.013066	0.6777	0.729
21.4	762	10	0.6936	0.013216	0.6682	0.720
21.5	752	2	0.6918	0.013245	0.6663	0.718
21.6	750	9	0.6835	0.013372	0.6578	0.710
21.7	741	4	0.6798	0.013427	0.6540	0.707
21.8	737	8	0.6724	0.013532	0.6464	0.699
21.8	729	6	0.6669	0.013608	0.6407	0.694
21.9	723	5	0.6623	0.013669	0.6360	0.690
22.0	718	9	0.6540	0.013775	0.6275	0.682
22.1	676	5	0.6491	0.013842	0.6225	0.677
22.2	671	5	0.6443	0.013907	0.6176	0.672
22.2	666	7	0.6375	0.013994	0.6107	0.666
22.3	659	5	0.6327	0.014054	0.6057	0.661
22.4	654	9	0.6240	0.014157	0.5968	0.652
22.5	645	3	0.6211	0.014190	0.5939	0.650
22.6	642	5	0.6162	0.014244	0.5889	0.645
22.7	637	5	0.6114	0.014295	0.5840	0.640
22.8	632	2	0.6095	0.014315	0.5820	0.638
22.8	630	7	0.6027	0.014383	0.5751	0.632
22.9	623	3	0.5998	0.014411	0.5722	0.629
23.0	620	5	0.5949	0.014457	0.5673	0.624
23.1	587	5	0.5899	0.014510	0.5621	0.619
23.2	582	6	0.5838	0.014571	0.5559	0.613
23.2	576	6	0.5777	0.014629	0.5497	0.607

23.3	570	10	0.5676	0.014720	0.5394	0.597
23.4	560	6	0.5615	0.014770	0.5333	0.591
23.5	554	7	0.5544	0.014825	0.5261	0.584
23.6	547	3	0.5514	0.014847	0.5230	0.581
23.7	544	4	0.5473	0.014875	0.5189	0.577
23.8	540	1	0.5463	0.014882	0.5179	0.576
23.8	539	3	0.5433	0.014903	0.5148	0.573
23.9	536	2	0.5412	0.014916	0.5128	0.571
24.0	534	7	0.5341	0.014959	0.5056	0.564
24.1	508	3	0.5310	0.014982	0.5024	0.561
24.2	505	5	0.5257	0.015017	0.4971	0.556
24.2	500	9	0.5163	0.015074	0.4875	0.547
24.3	491	4	0.5121	0.015097	0.4833	0.543
24.4	487	4	0.5078	0.015119	0.4791	0.538
24.5	483	4	0.5036	0.015139	0.4748	0.534
24.6	479	5	0.4984	0.015163	0.4695	0.529
24.7	474	3	0.4952	0.015176	0.4664	0.526
24.8	471	3	0.4921	0.015188	0.4632	0.523
24.8	468	3	0.4889	0.015199	0.4600	0.520
24.9	465	6	0.4826	0.015220	0.4537	0.513
25.0	459	9	0.4731	0.015245	0.4442	0.504
25.1	432	8	0.4644	0.015274	0.4354	0.495
25.2	424	6	0.4578	0.015292	0.4288	0.489
25.2	418	5	0.4523	0.015304	0.4233	0.483
25.3	413	4	0.4480	0.015311	0.4189	0.479
25.4	409	3	0.4447	0.015316	0.4156	0.476
25.5	406	5	0.4392	0.015322	0.4102	0.470
25.6	401	7	0.4315	0.015326	0.4025	0.463
25.7	394	4	0.4271	0.015326	0.3981	0.458
25.8	390	1	0.4261	0.015326	0.3970	0.457
25.8	389	4	0.4217	0.015324	0.3927	0.453
25.9	385	3	0.4184	0.015322	0.3894	0.450
26.0	382	5	0.4129	0.015316	0.3840	0.444
26.1	358	4	0.4083	0.015317	0.3794	0.439
26.2	354	4	0.4037	0.015317	0.3748	0.435
26.2	350	3	0.4002	0.015315	0.3713	0.431
26.3	347	3	0.3968	0.015313	0.3679	0.428
26.4	344	2	0.3945	0.015310	0.3656	0.426
26.5	342	3	0.3910	0.015306	0.3621	0.422
26.6	339	2	0.3887	0.015302	0.3598	0.420
26.7	337	3	0.3852	0.015296	0.3564	0.416
26.8	334	3	0.3818	0.015288	0.3529	0.413
26.8	331	3	0.3783	0.015280	0.3495	0.409
26.9	328	2	0.3760	0.015273	0.3472	0.407
27.0	326	4	0.3714	0.015259	0.3427	0.403
27.1	314	3	0.3678	0.015250	0.3391	0.399
27.2	311	3	0.3643	0.015240	0.3356	0.395
27.2	308	2	0.3619	0.015233	0.3333	0.393
27.3	306	2	0.3596	0.015225	0.3309	0.391
27.4	304	3	0.3560	0.015212	0.3274	0.387
27.5	301	2	0.3536	0.015202	0.3251	0.385
27.6	299	3	0.3501	0.015187	0.3216	0.381
27.8	296	1	0.3489	0.015182	0.3204	0.380
28.0	295	4	0.3442	0.015159	0.3157	0.375

28.1	276	1	0.3429	0.015155	0.3145	0.374
28.2	275	4	0.3379	0.015139	0.3095	0.369
28.2	271	5	0.3317	0.015114	0.3034	0.363
28.3	266	4	0.3267	0.015091	0.2984	0.358
28.4	262	3	0.3230	0.015072	0.2948	0.354
28.6	259	2	0.3205	0.015059	0.2923	0.351
28.8	257	2	0.3180	0.015044	0.2898	0.349
28.8	255	2	0.3155	0.015029	0.2874	0.346
28.9	253	1	0.3143	0.015022	0.2862	0.345
29.0	252	2	0.3118	0.015005	0.2837	0.343
29.1	244	3	0.3079	0.014983	0.2799	0.339
29.2	241	1	0.3066	0.014976	0.2787	0.337
29.5	240	1	0.3054	0.014968	0.2774	0.336
29.6	239	2	0.3028	0.014951	0.2749	0.334
29.7	237	2	0.3003	0.014934	0.2724	0.331
29.8	235	4	0.2952	0.014896	0.2674	0.326
29.8	231	2	0.2926	0.014877	0.2648	0.323
30.0	229	3	0.2888	0.014845	0.2611	0.319
30.2	221	2	0.2861	0.014826	0.2585	0.317
30.2	219	2	0.2835	0.014805	0.2560	0.314
30.3	217	1	0.2822	0.014794	0.2547	0.313
30.4	216	1	0.2809	0.014783	0.2534	0.311
30.5	215	3	0.2770	0.014749	0.2496	0.307
30.7	212	2	0.2744	0.014725	0.2470	0.305
30.8	210	3	0.2705	0.014688	0.2432	0.301
30.8	207	2	0.2679	0.014662	0.2406	0.298
31.0	205	1	0.2665	0.014648	0.2393	0.297
31.1	196	3	0.2625	0.014612	0.2353	0.293
31.2	193	1	0.2611	0.014600	0.2340	0.291
31.2	192	1	0.2597	0.014587	0.2327	0.290
31.4	191	1	0.2584	0.014574	0.2313	0.289
31.5	190	2	0.2557	0.014547	0.2287	0.286
31.6	188	1	0.2543	0.014533	0.2274	0.284
32.0	187	6	0.2462	0.014443	0.2194	0.276
32.1	177	1	0.2448	0.014428	0.2181	0.275
32.2	176	1	0.2434	0.014413	0.2167	0.273
32.4	175	2	0.2406	0.014382	0.2140	0.270
32.5	173	1	0.2392	0.014366	0.2126	0.269
32.7	172	2	0.2364	0.014333	0.2099	0.266
32.8	170	2	0.2336	0.014299	0.2072	0.263
33.1	166	1	0.2322	0.014282	0.2059	0.262
33.3	165	2	0.2294	0.014247	0.2031	0.259
33.4	163	2	0.2266	0.014210	0.2004	0.256
33.5	161	1	0.2252	0.014191	0.1990	0.255
33.6	160	1	0.2238	0.014172	0.1977	0.253
33.7	159	1	0.2224	0.014153	0.1963	0.252
33.8	158	1	0.2210	0.014133	0.1949	0.250
33.9	157	2	0.2182	0.014093	0.1922	0.248
34.0	155	2	0.2153	0.014051	0.1895	0.245
34.1	149	3	0.2110	0.013989	0.1853	0.240
34.2	146	3	0.2067	0.013924	0.1811	0.236
34.2	143	2	0.2038	0.013878	0.1783	0.233
34.5	141	1	0.2023	0.013855	0.1769	0.231
34.8	140	1	0.2009	0.013831	0.1755	0.230

34.8	139	2	0.1980	0.013782	0.1727	0.227
35.0	137	1	0.1965	0.013757	0.1714	0.225
35.1	131	1	0.1950	0.013734	0.1699	0.224
35.2	130	1	0.1935	0.013710	0.1685	0.222
35.2	129	1	0.1920	0.013685	0.1670	0.221
35.3	128	1	0.1905	0.013660	0.1656	0.219
35.4	127	1	0.1890	0.013635	0.1641	0.218
35.5	126	1	0.1875	0.013609	0.1627	0.216
35.6	125	1	0.1860	0.013583	0.1612	0.215
35.8	124	1	0.1845	0.013556	0.1598	0.213
35.9	123	2	0.1815	0.013500	0.1569	0.210
36.0	121	2	0.1785	0.013443	0.1540	0.207
36.1	118	3	0.1740	0.013354	0.1497	0.202
36.2	115	1	0.1725	0.013323	0.1483	0.201
36.2	114	1	0.1710	0.013292	0.1468	0.199
36.4	113	2	0.1680	0.013228	0.1439	0.196
36.6	111	1	0.1664	0.013195	0.1425	0.194
36.8	110	2	0.1634	0.013128	0.1396	0.191
36.8	108	1	0.1619	0.013093	0.1382	0.190
36.9	107	2	0.1589	0.013022	0.1353	0.187
37.0	105	1	0.1574	0.012986	0.1339	0.185
37.2	102	2	0.1543	0.012913	0.1309	0.182
37.4	100	2	0.1512	0.012838	0.1280	0.179
37.5	98	1	0.1496	0.012799	0.1266	0.177
37.6	97	3	0.1450	0.012679	0.1222	0.172
37.9	94	1	0.1435	0.012638	0.1207	0.171
38.2	92	1	0.1419	0.012596	0.1193	0.169
38.5	91	1	0.1404	0.012554	0.1178	0.167
38.6	90	1	0.1388	0.012511	0.1163	0.166
38.8	89	1	0.1372	0.012467	0.1149	0.164
39.1	82	1	0.1356	0.012427	0.1133	0.162
39.2	81	1	0.1339	0.012386	0.1117	0.161
39.5	80	1	0.1322	0.012343	0.1101	0.159
39.6	79	1	0.1305	0.012300	0.1085	0.157
39.7	78	1	0.1289	0.012256	0.1070	0.155
39.8	77	1	0.1272	0.012210	0.1054	0.154
39.9	76	1	0.1255	0.012164	0.1038	0.152
40.0	75	1	0.1238	0.012116	0.1022	0.150
40.3	70	1	0.1221	0.012072	0.1006	0.148
40.6	69	1	0.1203	0.012026	0.0989	0.146
40.8	68	1	0.1185	0.011978	0.0972	0.145
41.2	64	1	0.1167	0.011933	0.0955	0.143
42.2	60	1	0.1147	0.011892	0.0937	0.141
42.3	59	1	0.1128	0.011848	0.0918	0.139
42.5	58	1	0.1109	0.011803	0.0900	0.137
43.5	55	1	0.1088	0.011759	0.0881	0.135
45.2	52	1	0.1067	0.011718	0.0861	0.132
45.3	51	1	0.1047	0.011673	0.0841	0.130
46.2	45	2	0.1000	0.011609	0.0797	0.126
47.9	38	1	0.0974	0.011597	0.0771	0.123
52.6	32	1	0.0943	0.011627	0.0741	0.120
53.5	30	1	0.0912	0.011657	0.0710	0.117
54.7	27	1	0.0878	0.011704	0.0676	0.114
59.6	21	1	0.0836	0.011870	0.0633	0.110

```
summary(fit2)
```

```
Call: survfit(formula = so.2 ~ region, data = datos)
```

```
201 observations deleted due to missingness
```

```
      region=1
time n.risk n.event survival  std.err lower 95% CI upper 95% CI
 9.0  1373      1  0.9993 0.000728    0.9978      1.000
12.1  1372      1  0.9985 0.001029    0.9965      1.000
12.8  1371      1  0.9978 0.001260    0.9953      1.000
12.9  1370      1  0.9971 0.001455    0.9942      1.000
13.3  1369      1  0.9964 0.001626    0.9932      1.000
13.5  1368      1  0.9956 0.001780    0.9921      0.999
13.8  1367      1  0.9949 0.001922    0.9911      0.999
13.8  1366      1  0.9942 0.002054    0.9902      0.998
14.0  1365      1  0.9934 0.002178    0.9892      0.998
14.1  1364      1  0.9927 0.002295    0.9882      0.997
14.2  1363      3  0.9905 0.002614    0.9854      0.996
14.3  1360      3  0.9883 0.002896    0.9827      0.994
14.4  1357      3  0.9862 0.003153    0.9800      0.992
14.6  1354      2  0.9847 0.003312    0.9782      0.991
14.7  1352      2  0.9832 0.003464    0.9765      0.990
14.8  1350      1  0.9825 0.003537    0.9756      0.989
15.0  1349      1  0.9818 0.003608    0.9747      0.989
15.2  1274      2  0.9803 0.003764    0.9729      0.988
15.2  1272      1  0.9795 0.003839    0.9720      0.987
15.4  1271      2  0.9779 0.003984    0.9702      0.986
15.5  1269      3  0.9756 0.004193    0.9674      0.984
15.6  1266      3  0.9733 0.004390    0.9647      0.982
15.7  1263      2  0.9718 0.004516    0.9630      0.981
15.8  1261      2  0.9702 0.004639    0.9612      0.979
15.8  1259      2  0.9687 0.004758    0.9594      0.978
15.9  1257      1  0.9679 0.004816    0.9585      0.977
16.0  1256      2  0.9664 0.004930    0.9568      0.976
16.1  1207      1  0.9656 0.004991    0.9558      0.975
16.2  1206      2  0.9640 0.005109    0.9540      0.974
16.2  1204      3  0.9616 0.005281    0.9513      0.972
16.3  1201      3  0.9592 0.005447    0.9486      0.970
16.4  1198      1  0.9584 0.005501    0.9477      0.969
16.5  1197      2  0.9568 0.005607    0.9458      0.968
16.6  1195      1  0.9560 0.005659    0.9449      0.967
16.7  1194      2  0.9544 0.005762    0.9431      0.966
16.8  1192      5  0.9504 0.006010    0.9387      0.962
16.8  1187      4  0.9472 0.006199    0.9351      0.959
16.9  1183      3  0.9448 0.006337    0.9324      0.957
17.0  1180      6  0.9400 0.006601    0.9271      0.953
17.1  1113      4  0.9366 0.006790    0.9234      0.950
17.2  1109      2  0.9349 0.006882    0.9215      0.948
17.2  1107      4  0.9315 0.007061    0.9178      0.945
17.3  1103      5  0.9273 0.007277    0.9131      0.942
17.4  1098      3  0.9248 0.007403    0.9104      0.939
17.5  1095      2  0.9231 0.007485    0.9085      0.938
17.6  1093      2  0.9214 0.007566    0.9067      0.936
17.7  1091      4  0.9180 0.007725    0.9030      0.933
```

17.8	1087	1	0.9172	0.007764	0.9021	0.932
17.8	1086	5	0.9129	0.007954	0.8975	0.929
17.9	1081	3	0.9104	0.008065	0.8947	0.926
18.0	1078	4	0.9070	0.008210	0.8911	0.923
18.1	1022	4	0.9035	0.008368	0.8872	0.920
18.2	1018	4	0.8999	0.008521	0.8834	0.917
18.2	1014	4	0.8964	0.008671	0.8795	0.914
18.3	1010	5	0.8919	0.008852	0.8748	0.909
18.4	1005	2	0.8902	0.008923	0.8728	0.908
18.5	1003	3	0.8875	0.009028	0.8700	0.905
18.6	1000	6	0.8822	0.009231	0.8643	0.900
18.7	994	1	0.8813	0.009265	0.8633	0.900
18.8	993	5	0.8768	0.009428	0.8586	0.896
18.8	988	4	0.8733	0.009556	0.8548	0.892
18.9	984	8	0.8662	0.009802	0.8472	0.886
19.0	976	3	0.8635	0.009892	0.8444	0.883
19.1	935	3	0.8608	0.009989	0.8414	0.881
19.2	932	2	0.8589	0.010052	0.8394	0.879
19.2	930	6	0.8534	0.010239	0.8335	0.874
19.3	924	10	0.8441	0.010536	0.8237	0.865
19.4	914	7	0.8377	0.010735	0.8169	0.859
19.5	907	5	0.8331	0.010873	0.8120	0.855
19.6	902	7	0.8266	0.011060	0.8052	0.849
19.7	895	6	0.8210	0.011214	0.7994	0.843
19.8	889	4	0.8174	0.011315	0.7955	0.840
19.8	885	8	0.8100	0.011510	0.7877	0.833
19.9	877	4	0.8063	0.011605	0.7838	0.829
20.0	873	10	0.7970	0.011834	0.7742	0.821
20.1	830	7	0.7903	0.012004	0.7671	0.814
20.2	823	9	0.7817	0.012213	0.7581	0.806
20.2	814	2	0.7798	0.012259	0.7561	0.804
20.3	812	4	0.7759	0.012348	0.7521	0.800
20.4	808	5	0.7711	0.012457	0.7471	0.796
20.5	803	6	0.7653	0.012584	0.7411	0.790
20.6	797	4	0.7615	0.012666	0.7371	0.787
20.7	793	9	0.7529	0.012846	0.7281	0.778
20.8	784	3	0.7500	0.012904	0.7251	0.776
20.8	781	7	0.7433	0.013036	0.7181	0.769
20.9	774	7	0.7365	0.013163	0.7112	0.763
21.0	767	4	0.7327	0.013234	0.7072	0.759
21.1	719	6	0.7266	0.013357	0.7009	0.753
21.2	713	6	0.7205	0.013476	0.6945	0.747
21.2	707	4	0.7164	0.013553	0.6903	0.743
21.3	703	6	0.7103	0.013665	0.6840	0.738
21.4	697	8	0.7021	0.013809	0.6756	0.730
21.5	689	2	0.7001	0.013844	0.6735	0.728
21.6	687	8	0.6919	0.013979	0.6651	0.720
21.7	679	4	0.6879	0.014045	0.6609	0.716
21.8	675	6	0.6817	0.014140	0.6546	0.710
21.8	669	6	0.6756	0.014232	0.6483	0.704
21.9	663	5	0.6705	0.014306	0.6431	0.699
22.0	658	9	0.6614	0.014433	0.6337	0.690
22.1	618	5	0.6560	0.014513	0.6282	0.685
22.2	613	5	0.6507	0.014591	0.6227	0.680

22.2	608	7	0.6432	0.014695	0.6150	0.673
22.3	601	5	0.6378	0.014766	0.6095	0.667
22.4	596	8	0.6293	0.014875	0.6008	0.659
22.5	588	3	0.6260	0.014914	0.5975	0.656
22.6	585	5	0.6207	0.014978	0.5920	0.651
22.7	580	5	0.6153	0.015038	0.5866	0.646
22.8	575	2	0.6132	0.015062	0.5844	0.643
22.8	573	7	0.6057	0.015142	0.5768	0.636
22.9	566	2	0.6036	0.015164	0.5746	0.634
23.0	564	5	0.5982	0.015217	0.5691	0.629
23.1	533	3	0.5949	0.015255	0.5657	0.626
23.2	530	6	0.5881	0.015328	0.5588	0.619
23.2	524	6	0.5814	0.015397	0.5520	0.612
23.3	518	10	0.5702	0.015504	0.5406	0.601
23.4	508	4	0.5657	0.015543	0.5360	0.597
23.5	504	7	0.5578	0.015608	0.5281	0.589
23.6	497	3	0.5545	0.015635	0.5246	0.586
23.7	494	4	0.5500	0.015668	0.5201	0.582
23.8	490	3	0.5466	0.015693	0.5167	0.578
23.9	487	2	0.5444	0.015708	0.5144	0.576
24.0	485	6	0.5376	0.015753	0.5076	0.569
24.1	460	3	0.5341	0.015780	0.5041	0.566
24.2	457	5	0.5283	0.015822	0.4981	0.560
24.2	452	8	0.5189	0.015883	0.4887	0.551
24.3	444	4	0.5142	0.015911	0.4840	0.546
24.4	440	4	0.5096	0.015937	0.4793	0.542
24.5	436	4	0.5049	0.015962	0.4746	0.537
24.6	432	5	0.4990	0.015989	0.4687	0.531
24.7	427	3	0.4955	0.016005	0.4651	0.528
24.8	424	3	0.4920	0.016019	0.4616	0.524
24.8	421	3	0.4885	0.016032	0.4581	0.521
24.9	418	4	0.4839	0.016048	0.4534	0.516
25.0	414	8	0.4745	0.016075	0.4440	0.507
25.1	388	8	0.4647	0.016111	0.4342	0.497
25.2	380	5	0.4586	0.016130	0.4281	0.491
25.2	375	5	0.4525	0.016145	0.4219	0.485
25.3	370	4	0.4476	0.016155	0.4170	0.480
25.4	366	3	0.4439	0.016161	0.4134	0.477
25.5	363	5	0.4378	0.016168	0.4072	0.471
25.6	358	7	0.4293	0.016172	0.3987	0.462
25.7	351	4	0.4244	0.016172	0.3938	0.457
25.8	347	1	0.4231	0.016171	0.3926	0.456
25.8	346	4	0.4182	0.016168	0.3877	0.451
25.9	342	3	0.4146	0.016165	0.3841	0.448
26.0	339	5	0.4085	0.016156	0.3780	0.441
26.1	318	4	0.4033	0.016155	0.3729	0.436
26.2	314	4	0.3982	0.016153	0.3678	0.431
26.2	310	2	0.3956	0.016150	0.3652	0.429
26.3	308	2	0.3931	0.016147	0.3626	0.426
26.4	306	1	0.3918	0.016145	0.3614	0.425
26.5	305	3	0.3879	0.016139	0.3575	0.421
26.6	302	2	0.3853	0.016134	0.3550	0.418
26.7	300	3	0.3815	0.016126	0.3512	0.414
26.8	297	3	0.3776	0.016115	0.3473	0.411

26.8	294	3	0.3738	0.016104	0.3435	0.407
26.9	291	2	0.3712	0.016095	0.3410	0.404
27.0	289	2	0.3686	0.016086	0.3384	0.402
27.1	280	2	0.3660	0.016079	0.3358	0.399
27.2	278	3	0.3621	0.016066	0.3319	0.395
27.2	275	2	0.3594	0.016057	0.3293	0.392
27.3	273	1	0.3581	0.016052	0.3280	0.391
27.4	272	3	0.3542	0.016036	0.3241	0.387
27.5	269	2	0.3515	0.016024	0.3215	0.384
27.6	267	3	0.3476	0.016006	0.3176	0.380
27.8	264	1	0.3463	0.015999	0.3163	0.379
28.0	263	3	0.3423	0.015978	0.3124	0.375
28.1	249	1	0.3409	0.015973	0.3110	0.374
28.2	248	4	0.3354	0.015950	0.3056	0.368
28.2	244	5	0.3286	0.015917	0.2988	0.361
28.3	239	4	0.3231	0.015886	0.2934	0.356
28.4	235	3	0.3189	0.015861	0.2893	0.352
28.6	232	2	0.3162	0.015843	0.2866	0.349
28.8	230	2	0.3134	0.015824	0.2839	0.346
28.8	228	2	0.3107	0.015804	0.2812	0.343
28.9	226	1	0.3093	0.015794	0.2799	0.342
29.0	225	1	0.3079	0.015784	0.2785	0.340
29.1	219	2	0.3051	0.015764	0.2757	0.338
29.2	217	1	0.3037	0.015754	0.2744	0.336
29.5	216	1	0.3023	0.015744	0.2730	0.335
29.7	215	2	0.2995	0.015722	0.2702	0.332
29.8	213	4	0.2939	0.015677	0.2647	0.326
29.8	209	2	0.2911	0.015652	0.2620	0.323
30.0	207	3	0.2869	0.015614	0.2578	0.319
30.2	199	2	0.2840	0.015589	0.2550	0.316
30.2	197	2	0.2811	0.015564	0.2522	0.313
30.3	195	1	0.2796	0.015551	0.2508	0.312
30.5	194	3	0.2753	0.015509	0.2465	0.307
30.7	191	1	0.2739	0.015495	0.2451	0.306
30.8	190	3	0.2696	0.015450	0.2409	0.302
30.8	187	2	0.2667	0.015419	0.2381	0.299
31.0	185	1	0.2652	0.015403	0.2367	0.297
31.1	176	3	0.2607	0.015360	0.2323	0.293
31.2	173	1	0.2592	0.015345	0.2308	0.291
31.2	172	1	0.2577	0.015329	0.2293	0.290
31.4	171	1	0.2562	0.015313	0.2279	0.288
31.5	170	2	0.2532	0.015281	0.2249	0.285
31.6	168	1	0.2517	0.015264	0.2235	0.283
32.0	167	4	0.2456	0.015193	0.2176	0.277
32.2	159	1	0.2441	0.015176	0.2161	0.276
32.4	158	2	0.2410	0.015140	0.2131	0.273
32.5	156	1	0.2395	0.015122	0.2116	0.271
32.7	155	2	0.2364	0.015084	0.2086	0.268
32.8	153	1	0.2348	0.015064	0.2071	0.266
33.1	150	1	0.2333	0.015045	0.2056	0.265
33.3	149	2	0.2301	0.015005	0.2025	0.262
33.4	147	2	0.2270	0.014963	0.1995	0.258
33.5	145	1	0.2254	0.014942	0.1980	0.257
33.6	144	1	0.2239	0.014920	0.1965	0.255

33.7	143	1	0.2223	0.014897	0.1949	0.254
33.8	142	1	0.2207	0.014874	0.1934	0.252
33.9	141	2	0.2176	0.014827	0.1904	0.249
34.0	139	2	0.2145	0.014778	0.1874	0.245
34.1	133	2	0.2112	0.014731	0.1843	0.242
34.2	131	3	0.2064	0.014656	0.1796	0.237
34.2	128	2	0.2032	0.014603	0.1765	0.234
34.5	126	1	0.2016	0.014576	0.1749	0.232
34.8	125	1	0.2000	0.014548	0.1734	0.231
34.8	124	2	0.1967	0.014491	0.1703	0.227
35.0	122	1	0.1951	0.014462	0.1687	0.226
35.1	116	1	0.1934	0.014435	0.1671	0.224
35.2	115	1	0.1918	0.014407	0.1655	0.222
35.3	114	1	0.1901	0.014379	0.1639	0.220
35.4	113	1	0.1884	0.014349	0.1623	0.219
35.5	112	1	0.1867	0.014320	0.1607	0.217
35.6	111	1	0.1850	0.014289	0.1590	0.215
35.8	110	1	0.1833	0.014258	0.1574	0.214
35.9	109	2	0.1800	0.014193	0.1542	0.210
36.0	107	2	0.1766	0.014126	0.1510	0.207
36.1	104	3	0.1715	0.014021	0.1461	0.201
36.2	101	1	0.1698	0.013985	0.1445	0.200
36.2	100	1	0.1681	0.013948	0.1429	0.198
36.4	99	1	0.1664	0.013910	0.1413	0.196
36.8	98	2	0.1630	0.013832	0.1381	0.193
36.8	96	1	0.1613	0.013792	0.1364	0.191
36.9	95	2	0.1579	0.013709	0.1332	0.187
37.0	93	1	0.1562	0.013666	0.1316	0.185
37.2	90	2	0.1528	0.013581	0.1283	0.182
37.4	88	2	0.1493	0.013493	0.1251	0.178
37.6	86	3	0.1441	0.013353	0.1202	0.173
37.9	83	1	0.1424	0.013304	0.1185	0.171
38.2	82	1	0.1406	0.013255	0.1169	0.169
38.5	81	1	0.1389	0.013204	0.1153	0.167
38.6	80	1	0.1371	0.013153	0.1136	0.166
38.8	79	1	0.1354	0.013101	0.1120	0.164
39.1	73	1	0.1336	0.013052	0.1103	0.162
39.2	72	1	0.1317	0.013002	0.1085	0.160
39.5	71	1	0.1298	0.012950	0.1068	0.158
39.6	70	1	0.1280	0.012897	0.1051	0.156
39.7	69	1	0.1261	0.012843	0.1033	0.154
39.8	68	1	0.1243	0.012787	0.1016	0.152
39.9	67	1	0.1224	0.012730	0.0999	0.150
40.0	66	1	0.1206	0.012672	0.0981	0.148
40.3	61	1	0.1186	0.012617	0.0963	0.146
40.8	60	1	0.1166	0.012561	0.0944	0.144
41.2	58	1	0.1146	0.012504	0.0925	0.142
42.2	56	1	0.1126	0.012447	0.0906	0.140
42.3	55	1	0.1105	0.012388	0.0887	0.138
42.5	54	1	0.1085	0.012327	0.0868	0.136
43.5	51	1	0.1063	0.012267	0.0848	0.133
45.3	48	1	0.1041	0.012210	0.0827	0.131
46.2	43	2	0.0993	0.012113	0.0782	0.126
47.9	37	1	0.0966	0.012079	0.0756	0.123

52.6	31	1	0.0935	0.012084	0.0726	0.120
53.5	29	1	0.0903	0.012090	0.0694	0.117
54.7	26	1	0.0868	0.012113	0.0660	0.114
59.6	20	1	0.0824	0.012260	0.0616	0.110

region=8						
time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI
12.0	155	1	0.994	0.00643	0.9810	1.000
13.2	154	1	0.987	0.00906	0.9695	1.000
13.9	153	1	0.981	0.01107	0.9592	1.000
14.6	152	1	0.974	0.01274	0.9495	0.999
14.8	151	1	0.968	0.01419	0.9403	0.996
15.0	150	2	0.955	0.01668	0.9227	0.988
15.3	136	1	0.948	0.01797	0.9132	0.984
15.5	135	1	0.941	0.01916	0.9040	0.979
15.8	134	1	0.934	0.02027	0.8949	0.974
15.9	133	1	0.927	0.02129	0.8859	0.969
16.1	128	1	0.920	0.02233	0.8768	0.964
16.3	127	1	0.912	0.02329	0.8677	0.959
16.5	126	1	0.905	0.02421	0.8588	0.954
16.6	125	1	0.898	0.02507	0.8500	0.948
16.8	124	1	0.891	0.02590	0.8412	0.943
17.0	123	2	0.876	0.02742	0.8239	0.932
17.1	116	1	0.869	0.02821	0.8150	0.926
17.2	115	1	0.861	0.02896	0.8060	0.920
17.3	114	1	0.853	0.02967	0.7972	0.914
17.5	113	1	0.846	0.03035	0.7884	0.908
18.2	103	1	0.838	0.03115	0.7788	0.901
18.2	102	1	0.829	0.03191	0.7692	0.894
18.5	101	1	0.821	0.03263	0.7597	0.888
18.7	100	1	0.813	0.03332	0.7503	0.881
18.8	99	1	0.805	0.03398	0.7409	0.874
18.9	98	1	0.797	0.03462	0.7316	0.867
19.0	97	1	0.788	0.03522	0.7223	0.861
19.2	89	1	0.780	0.03592	0.7122	0.853
19.3	88	1	0.771	0.03659	0.7022	0.846
19.6	87	1	0.762	0.03722	0.6922	0.838
19.8	86	1	0.753	0.03783	0.6823	0.831
19.8	85	1	0.744	0.03841	0.6725	0.823
19.9	84	1	0.735	0.03896	0.6627	0.816
20.2	81	2	0.717	0.04006	0.6427	0.800
20.6	79	3	0.690	0.04151	0.6131	0.776
20.7	76	2	0.672	0.04235	0.5936	0.760
20.8	74	2	0.654	0.04311	0.5743	0.744
21.0	72	2	0.635	0.04378	0.5551	0.727
21.4	65	2	0.616	0.04456	0.5344	0.710
21.6	63	1	0.606	0.04492	0.5241	0.701
21.8	62	2	0.587	0.04554	0.5037	0.683
22.4	58	1	0.576	0.04587	0.4932	0.674
22.9	57	1	0.566	0.04616	0.4827	0.664
23.1	54	2	0.545	0.04678	0.4609	0.645
23.4	52	2	0.524	0.04727	0.4394	0.626
23.8	50	1	0.514	0.04747	0.4287	0.616
24.0	49	1	0.503	0.04765	0.4181	0.606

24.2	48	1	0.493	0.04780	0.4076	0.596
24.9	47	2	0.472	0.04801	0.3866	0.576
25.0	45	1	0.461	0.04807	0.3762	0.566
25.2	44	1	0.451	0.04811	0.3658	0.556
26.2	40	1	0.440	0.04821	0.3546	0.545
26.3	39	1	0.428	0.04827	0.3435	0.534
26.4	38	1	0.417	0.04830	0.3324	0.523
27.0	37	2	0.395	0.04825	0.3105	0.501
27.1	34	1	0.383	0.04821	0.2992	0.490
27.3	33	1	0.371	0.04812	0.2881	0.479
28.0	32	1	0.360	0.04800	0.2770	0.467
29.0	27	1	0.346	0.04803	0.2640	0.455
29.1	25	1	0.333	0.04807	0.2505	0.441
29.6	24	2	0.305	0.04789	0.2241	0.415
30.4	22	1	0.291	0.04768	0.2111	0.401
30.7	21	1	0.277	0.04738	0.1982	0.387
32.0	20	2	0.249	0.04652	0.1731	0.359
32.1	18	1	0.236	0.04595	0.1607	0.345
32.8	17	1	0.222	0.04529	0.1486	0.331
34.1	16	1	0.208	0.04453	0.1366	0.316
35.2	15	1	0.194	0.04366	0.1248	0.302
36.4	14	1	0.180	0.04269	0.1132	0.287
36.6	13	1	0.166	0.04159	0.1018	0.271
37.5	12	1	0.152	0.04037	0.0907	0.256
40.6	9	1	0.135	0.03927	0.0768	0.239
45.2	4	1	0.102	0.04157	0.0456	0.227

a.

La edad mediana de la primera unión es de aproximadamente 24 años para las personas de las regiones Metropolitana y Pacífico Central

La edad mediana de la primera unión es de aproximadamente 24,6 años para las personas de la región Metropolitana

La edad mediana de la primera unión es de aproximadamente 24 años para las personas de la región Pacífico Central

b.

Aproximadamente el 25 % de las personas de las regiones Pacífico Central y Metropolitana habían experimentado su primera unión conyugal al alcanzar los 20,65 años

Aproximadamente el 25 % de las personas de la región Pacífico Central había experimentado su primera unión conyugal al alcanzar los 20,8 años

Aproximadamente el 25 % de las personas de la región Pacífico Central había experimentado su primera unión conyugal al alcanzar los 19,8 años

c. Cree una curva de hazards

Pacífico Central y Metropolitana juntos

```
tabla_hazard <- data.frame(
  Tiempo = fit1$time,
  Sobrevivientes = fit1$n.risk,
  Eventos = fit1$n.event
) |>
```

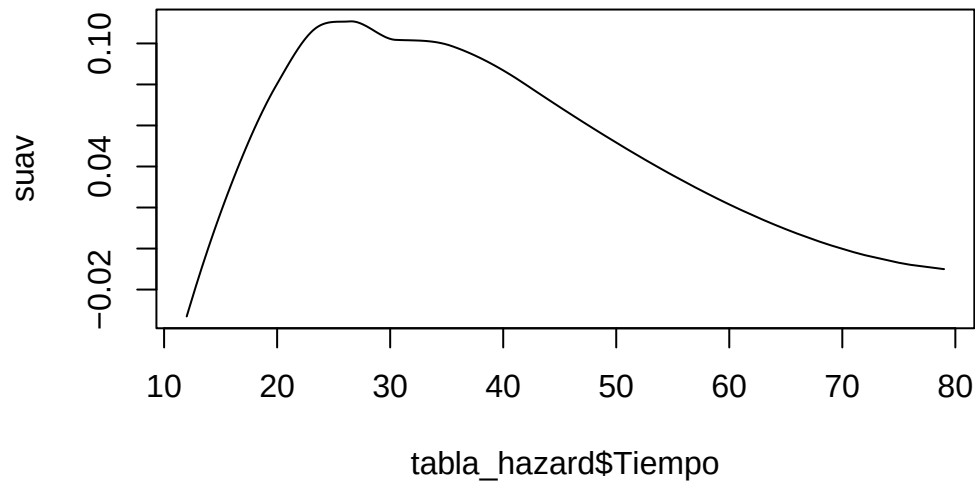
```

mutate(
  delta_t = Tiempo - lag(Tiempo),
  ht = (Eventos / Sobrevivientes) / delta_t
) |>
filter(is.finite(ht))

suavizam <- loess(ht ~ Tiempo, data = tabla_hazard)
suav <- predict(suavizam)

plot(
  suav ~ tabla_hazard$Tiempo,
  type = "l",
)

```



Pacífico Central y Metropolitana separadas

```

tabla_hazard <- data.frame(
  Tiempo = fit2$time,
  Sobrevivientes = fit2$n.risk,
  Eventos = fit2$n.event,
  region = rep(names(fit2$strata), fit2$strata)
) |>
group_by(region) |>
mutate(
  delta_t = Tiempo - lag(Tiempo),
  ht = (Eventos / Sobrevivientes) / delta_t
) |>
filter(is.finite(ht))

suavizam <- tabla_hazard |>
group_by(region) |>

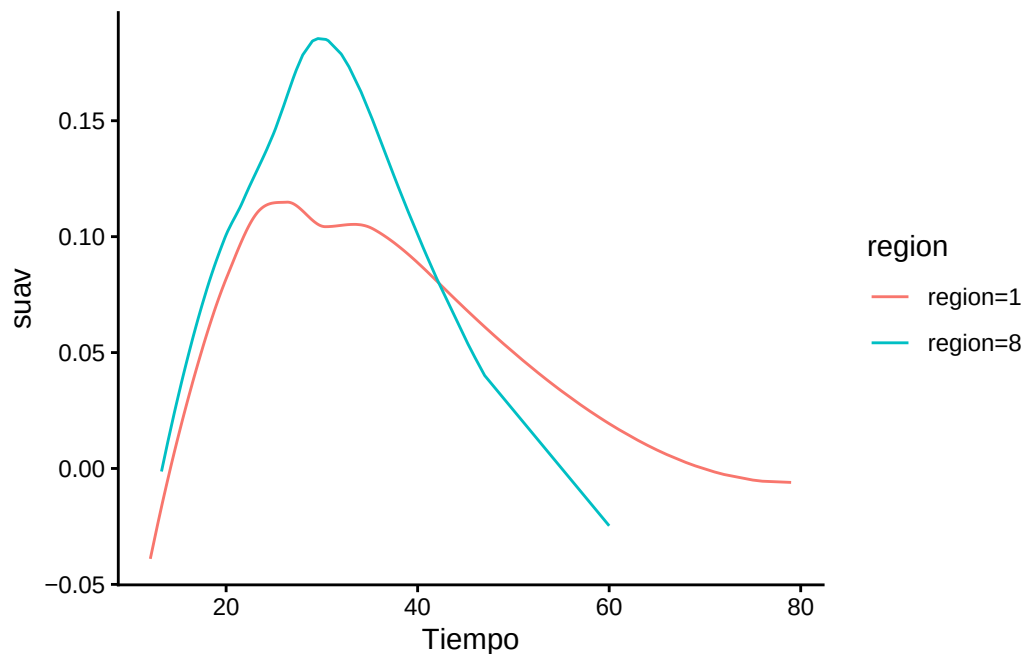
```

```

mutate(suav = predict(loess(ht ~ Tiempo)))

ggplot(
  suavizam,
  aes(Tiempo, suav, color = region)
) +
  geom_line() +
  theme_classic()

```



Para la región Pacífico Central, hasta la edad de 45 años aproximadamente, los hazards de la primera unión son mayores que para la región Metropolitana, sin embargo, a partir de esta edad es al revés. Esto quiere decir que en la región Metropolitana, las personas que nunca se han unido conyugalmente tienen menor propensión de unirse comparado con los de la región Pacífico Central hasta los 45 años. En otras palabras, las personas que no se han unido en la región Metropolitana tienden a hacerlo a mayor edad.

Ejercicio 4

```

datos <- datos |>
mutate(
  ednivel_cat = case_when(
    ednivel <= 10 ~ "Primaria o menos",
    ednivel <= 30 ~ "Secundaria",
    ednivel <= 50 ~ "Postsecundaria"
  )
)

```

i Nota

El `case_when` ya hace automáticamente el NA con los casos fuera de rango.
Se verificó con `: table(datos$ednivel_cat, useNA = "ifany")`
PD: Aplica para antes y después de esto también.

Ejercicio 5

```
datos <- datos |>
mutate(
  religion_cat = case_when(
    r209 == 1 ~ "Católica",
    r209 %in% c(2, 3, 4, 5, 7) ~ "Otra religión",
    r209 == 8 ~ "No creyente"
  ),
  sexo_cat = factor(
    r204,
    levels = c(1, 2),
    labels = c("Hombre", "Mujer")
  ),
  zona_cat = case_when(
    zona == 1 ~ "Urbana",
    zona == 2 ~ "Rural"
  )
)
```

Ejercicio 6

Regiones unidas

```
fit.sexo <- survfit(so.1 ~ as.factor(sexo_cat), data = datos)
fit.educacion <- survfit(so.1 ~ as.factor(ednivel_cat), data = datos)
fit.religion <- survfit(so.1 ~ as.factor(religion_cat), data = datos)
fit.zona <- survfit(so.1 ~ as.factor(zona_cat), data = datos)
```

```
graf.sexo <- ggsurvplot(
  fit.sexo,
  data = datos,
  pval = TRUE,
  title = "Sexo",
  legend.labs = c("Hombre", "Mujer")
)

graf.educacion <- ggsurvplot(
  fit.educacion,
  data = datos,
  pval = TRUE,
  title = "Educación",
  legend.labs = c("Postsecundaria", "Primaria o menos", "Secundaria")
)
```

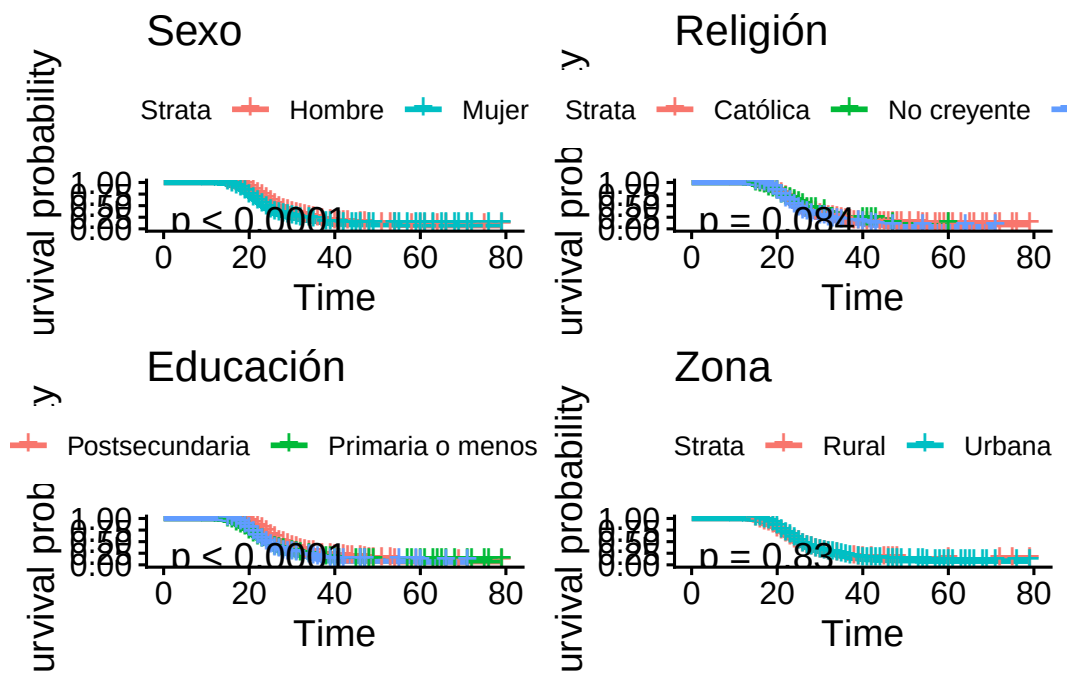
```

graf.religion <- ggsurvplot(
  fit.religion,
  data = datos,
  pval = TRUE,
  title = "Religión",
  legend.labs = c("Católica", "No creyente", "Otra")
)

graf.zona <- ggsurvplot(
  fit.zona,
  data = datos,
  pval = TRUE,
  title = "Zona",
  legend.labs = c("Rural", "Urbana")
)

arrange_ggsurvplots(
  list(
    graf.sexo ,
    graf.educacion ,
    graf.religion ,
    graf.zona
  ),
  ncol = 2,
  nrow = 2,
  print = TRUE
)

```



```

datos <- datos |>
tidyr::drop_na(ednivel_cat) |>

```

```
mutate(ednivel_cat = factor(ednivel_cat))

pairwise_survdif(Surv(time = tiempo_union, event = evento) ~ ednivel_cat, data = datos, p.adjust.me
```

Pairwise comparisons using Log-Rank test

data: datos and ednivel_cat

	Postsecundaria	Primaria o menos
Primaria o menos	5.4e-07	-
Secundaria	1.7e-09	0.99

P value adjustment method: holm

💡 Interpretación

De acuerdo con las pruebas log-rank, con una significancia del 5 %:

H_0 : Todos los grupos tienen la misma curva de supervivencia

H_1 : Al menos uno de los grupos tiene una curva de supervivencia diferente

Para la zona no hay diferencias significativas respecto a las curvas de supervivencia ($p = 0,83$)

Para el sexo sí existen diferencias significativas respecto a las curvas de supervivencia ($p < 0,0001$)

Para religión no hay diferencias significativas respecto a las curvas de supervivencia ($p = 0,084$)

Para educación sí existen diferencias significativas respecto a las curvas de supervivencia ($p < 0,0001$)

Específicamente para educación existen diferencias en las curvas de supervivencia entre Primaria o menos y Postsecundaria ($p < 0,0001$) así como entre Secundaria y Postsecundaria ($p < 0,0001$). Entre Primaria o menos y Secundaria no existen diferencias significativas ($p = 0,99$)

Por región

```
datos <- as.data.frame(datos) # Esto se hizo porque los labels de STATA daban problemas con el facet
datos <- zap_labels(datos)
datos <- zap_formats(datos)
datos$sexo_cat <- as.factor(datos$sexo_cat)
datos$ednivel_cat <- as.factor(datos$ednivel_cat)
datos$religion_cat <- as.factor(datos$religion_cat)
datos$zona_cat <- as.factor(datos$zona_cat)
```

```
fit.sexo <- survfit(Surv(time = tiempo_union, event = evento) ~ sexo_cat, data = datos)
fit.educacion <- survfit(Surv(time = tiempo_union, event = evento) ~ ednivel_cat, data = datos)
fit.religion <- survfit(Surv(time = tiempo_union, event = evento) ~ religion_cat, data = datos)
fit.zona <- survfit(Surv(time = tiempo_union, event = evento) ~ zona_cat, data = datos)
```

```
graf.sexo <- ggsurvplot_facet(
  fit.sexo,
  data = datos,
  facet.by = "region",
  pval = TRUE,
  title = "Sexo"
)
```

```

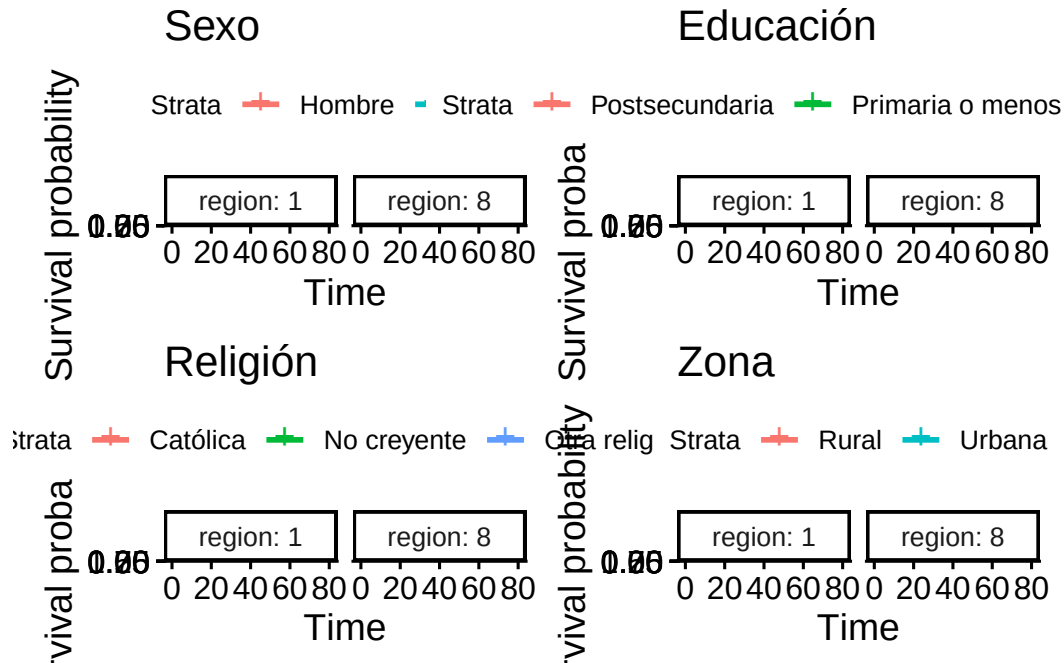
graf.educacion <- ggsurvplot_facet(
  fit.educacion,
  data = datos,
  facet.by = "region",
  pval = TRUE,
  title = "Educación"
)

graf.religion <- ggsurvplot_facet(
  fit.religion,
  data = datos,
  facet.by = "region",
  pval = TRUE,
  title = "Religión"
)

graf.zona <- ggsurvplot_facet(
  fit.zona,
  data = datos,
  facet.by = "region",
  pval = TRUE,
  title = "Zona"
)

patchwork::wrap_plots(
  graf.sexo,
  graf.educacion,
  graf.religion,
  graf.zona
,
  ncol = 2
)

```



```
pairwise_survdif(Surv(time = tiempo_union, event = evento) ~ ednivel_cat , data = datos[datos$region == "1", ])
```

Pairwise comparisons using Log-Rank test

```
data: datos[datos$region == "1", ] and ednivel_cat
```

	Postsecundaria	Primaria o menos
Primaria o menos	3.9e-07	-
Secundaria	9.7e-11	0.72

P value adjustment method: BH

💡 Interpretación

De acuerdo con las pruebas log-rank, con una significancia del 5 %:

H_0 : Todos los grupos tienen la misma curva de supervivencia

H_1 : Al menos uno de los grupos tiene una curva de supervivencia diferente

Para la zona no hay diferencias significativas respecto a las curvas de supervivencia dentro la región Metropolitana ni para la Pacífico Central ($p = 0,59$; $p = 0,69$)

Para el sexo sí existen diferencias significativas respecto a las curvas de supervivencia dentro la región Metropolitana y dentro de la Pacífico Central ($p < 0,0001$; $p = 0,0076$).

Para religión no hay diferencias significativas respecto a las curvas de supervivencia dentro la región Metropolitana ni dentro la Pacífico Central ($p = 0,2$; $p = 0,41$)

En el caso de educación sí existen diferencias significativas respecto a las curvas de supervivencia para la región Metropolitana ($p < 0,0001$) pero no así para la región Pacífico Central ($p = 0,69$).

Específicamente para educación en la región Metropolitana existen diferencias en las curvas de supervivencia entre Primaria o menos y Postsecundaria ($p < 0,0001$) así como entre Secundaria y Postsecundaria ($p < 0,0001$). Entre Primaria o menos y Secundaria no existen diferencias significativas ($p = 0,72$).

Ejercicio 7

Para todas las pruebas se utiliza un nivel de significancia de $\alpha = 0.05$.

Sexo

```
prueba.sexo <- survdiff(Surv(time = tiempo_union, event = evento) ~ sexo_cat, data = datos)
prueba.sexo$pvalue
```

[1] 4.0801e-08

Hipótesis nula: $H_0 : S_{Hombre}(t) = S_{Mujer}(t) \Rightarrow h_{Hombre}(t) = h_{Mujer}(t)$.

Hipótesis alternativa: $H_1 : S_{Hombre}(t) \neq S_{Mujer}(t) \Rightarrow h_{Hombre}(t) \neq h_{Mujer}(t)$.

Interpretación: Como el p-value es menor 0.0001, y por tanto, menor que 0,05, se rechaza la hipótesis nula. Hay evidencia estadísticamente significativa de que las curvas de supervivencia de la edad a la primera unión son diferentes entre hombres y mujeres considerando ambas regiones.

Educación

```
prueba.educacion <- survdiff(Surv(time = tiempo_union, event = evento) ~ ednivel_cat, data = datos)
prueba.educacion$pvalue
```

[1] 3.629024e-09

Hipótesis nula: $H_0 : S_{Primaria}(t) = S_{Secundaria}(t) = S_{Postsecundaria}(t) \Rightarrow h_{Primaria}(t) = h_{Secundaria}(t) = h_{Postsecundaria}(t)$.

Hipótesis alternativa: $H_1 : \text{al menos una } S_i(t) \neq S_j(t) \Rightarrow \text{al menos una } h_i(t) \neq h_j(t)$.

Interpretación: Como el p-value es menor que 0,001, y por tanto menor que 0,05, se rechaza la hipótesis nula. Hay evidencia estadísticamente significativa de diferencias entre las curvas según el nivel educativo considerando ambas regiones.

Religión

```
prueba.religion <- survdiff(Surv(time = tiempo_union, event = evento) ~ religion_cat, data = datos)
prueba.religion$pvalue
```

[1] 0.08153174

Hipótesis nula: $H_0 : S_{Catlica}(t) = S_{Otra}(t) = S_{NoCreyente}(t) \Rightarrow h_{Catlica}(t) = h_{Otra}(t) = h_{NoCreyente}(t)$.

Hipótesis alternativa: $H_1 : \text{al menos una } S_i(t) \neq S_j(t) \Rightarrow \text{al menos una } h_i(t) \neq h_j(t)$.

Interpretación: Como el p-value es 0,0815, mayor que 0,05, no se rechaza la hipótesis nula. No hay evidencia estadísticamente significativa de diferencias entre las curvas según la religión considerando ambas regiones.

Zona

```
prueba.zona <- survdiff(Surv(time = tiempo_union, event = evento) ~ zona_cat, data = datos)
prueba.zona$pvalue
```

[1] 0.8160423

Hipótesis nula: $H_0 : S_{Urbana}(t) = S_{Rural}(t) \Rightarrow h_{Urbana}(t) = h_{Rural}(t)$.

Hipótesis alternativa: $H_1 : S_{Urbana}(t) \neq S_{Rural}(t) \Rightarrow h_{Urbana}(t) \neq h_{Rural}(t)$.

Interpretación: Como el p-value es 0.816, mayor que 0,05, no se rechaza la hipótesis nula. No hay evidencia estadísticamente significativa de diferencias entre las curvas de las zonas urbana y rural considerando ambas regiones.

Ejercicio 8

En `survdif()`, `rho = 0` corresponde a la prueba log-rank y `rho = 1` corresponde a la prueba Peto-Peto.

```
survdif(  
  Surv(time = tiempo_union, event = evento) ~ sexo_cat,  
  data = datos,  
  rho = 1  
)$pvalue
```

```
[1] 3.803148e-14
```

i Interpretación

La prueba log-rank produce un p-value menor que 0,0001 y la prueba Peto-Peto un p-value menor que 0,0001. Ambos valores son menores que 0,05, por lo que con ambas pruebas se rechaza la hipótesis nula de igualdad entre las curvas de supervivencia de hombres y mujeres considerando ambas regiones. Por tanto, la prueba Peto-Peto conduce a la misma conclusión que la prueba log-rank, es decir, existen diferencias estadísticamente significativas por sexo en la edad a la primera unión conyugal considerando ambas regiones.